MT 1003 – Calculus and Analytical Geometry

Homework #7

Coordinate Systems: Conversions, Applications and Higher Dimensions

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Instructions:

1. Name the submitted file as `Section_RollNumber_HW7` (for example, `A_26I1234_HW7`).
2. Submit the complete homework as one scanned PDF file.
3. Show the main conversion steps. Answer conceptual parts in two or three sentences, using a small sketch where helpful.
4. All angles are in radians unless degrees are explicitly specified. Give exact answers wherever possible.
5. For Section B, include the Python code, generated graphs or numerical output, and your written observations in the same PDF file.

Section A: Analytical Work

Conventions. Cylindrical coordinates are (r, θ, z) and spherical coordinates are (ρ, θ, ϕ), where ϕ is measured from the positive z-axis. Use $r, \rho \geq 0$, $0 \leq \theta < 2\pi$ and $0 \leq \phi \leq \pi$.

$$x = r\cos\theta, \quad y = r\sin\theta; \qquad r = \rho\sin\phi, \quad z = \rho\cos\phi, \quad \rho^2 = r^2 + z^2.$$

Q1. A short conversion exercise.

Convert each point to the other two three-dimensional coordinate systems. Determine the angle θ from the signs of x and y. Give exact values.

(a) Rectangular: $(x, y, z) = (1, 1, \sqrt{2})$.

(b) Cylindrical: $(r, \theta, z) = (2, 5\pi/6, -2)$.

(c) Spherical: $(\rho, \theta, \phi) = (4, \pi/3, \pi/6)$.

Q2. What do the coordinates mean?

(a) Explain the difference between r and ρ. When are they equal?

(b) For a point on the positive z-axis, which spherical angle is fixed and which is nonunique? What changes at the origin?

(c) A student uses $\theta = \arctan(y/x)$ for $(-1, 1, 2)$ and obtains $-\pi/4$. Explain the mistake and give an angle in the required range.

Q3. Geometry and choice of coordinates.

(a) Identify $r = 3$, $\rho = 3$ and $\phi = \pi/3$ geometrically. Why do they describe different surfaces?

(b) Which coordinates would you choose for a straight circular pipe, and which for points at a fixed distance from a sensor? Explain your choices.

(c) Does converting coordinates move a point or change the distance between two points? Explain why differences of angular coordinates alone do not give Euclidean distance.

Q4. Beyond three dimensions.

In $\mathbb{R}^n$ ($n \geq 2$), spherical-type coordinates use one radius and $n-1$ angles. A cylindrical-type construction uses polar coordinates in two components and leaves the remaining components rectangular. These descriptions have coordinate singularities where some angles are nonunique.

(a) What is the two-dimensional analogue of spherical coordinates? Explain how adding one extra rectangular coordinate produces cylindrical coordinates in $\mathbb{R}^3$.

(b) In $\mathbb{R}^5$, how many angles accompany one spherical radius? For the cylindrical-type representation $(r, \theta, x_3, x_4, x_5)$, explain why $r = \sqrt{x_1^2 + x_2^2}$ is generally not distance from the origin.

(c) A program plots only (x_1, x_2, x_3) of a five-dimensional vector. Can different vectors appear at the same plotted point? Give an example and explain what information is lost.

Q5. Rectangular, oblique and curvilinear coordinates.

Coordinate curves are obtained by varying one coordinate while fixing the others. Curvilinear coordinates allow curved coordinate lines; straight coordinate lines need not be perpendicular.

(a) For $x = u + v$, $y = v$, sketch the lines $u = 0$ and $v = 0$. Explain why the coordinates are oblique rather than rectangular.

(b) For $x = u$, $y = v + u^2$, identify the curves $u = c$ and $v = c$. Must every coordinate curve be curved for a system to be curvilinear?

(c) Is "rectangular or curved" a complete classification of coordinate grids? Use the preceding examples to justify your answer.

Section B: A Short Python Investigation

Q6. Choosing the angle in the xy-plane.

Run the code. The function `arctan2(y,x)` uses the signs of x and y to determine the angle in the xy-plane.

```
import numpy as np
points = [(1, 1), (-1, 1), (-1, -1), (1, -1)]
for x, y in points:
    naive = np.arctan(y/x)
    theta = np.mod(np.arctan2(y, x), 2*np.pi)
    print((x, y), "arctan:", round(naive, 3),
          "atan2:", round(theta, 3))
```

(a) Which points give the wrong direction with $\arctan(y/x)$? Distinguish a wrong direction from a negative angle representing the correct direction.

(b) Add $(0, 2)$ to the list and remove `naive` from both the calculation and the print statement. Explain why `arctan2` can handle this point while the ratio y/x cannot.